Coordinate Geometry

The big idea

Coordinate geometry turns shapes into numbers. Once points have coordinates, we can find lengths, check whether points are collinear and locate points that divide a segment, all with formulas.

Distance formula

DISTANCE BETWEEN $P(X_1, Y_1)$ AND $Q(X_2, Y_2)$

$$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

It is just Pythagoras: the horizontal gap and the vertical gap are the legs of a right triangle. The distance of P(x, y) from the origin is $\sqrt{(x^2 + y^2)}$.

ARE (1, 5), (2, 3) AND (−2, −11) COLLINEAR?

AB = $\sqrt{(1 + 4)} = \sqrt{5}$, BC = $\sqrt{(16 + 196)} = \sqrt{212}$, AC = $\sqrt{(9 + 256)} = \sqrt{265}$.
$\sqrt{5} + \sqrt{212} \neq \sqrt{265}$, so the points are **not collinear**.

PROVING A SHAPE

Equal four sides and equal diagonals ⇒ square. Equal opposite sides and equal diagonals ⇒ rectangle. Equal four sides, unequal diagonals ⇒ rhombus.

Section formula

POINT DIVIDING AB INTERNALLY IN THE RATIO M : N

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

MIDPOINT (RATIO 1 : 1)

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

FIND THE POINT DIVIDING (4, −3) AND (8, 5) IN THE RATIO 3 : 1

$x = (3 \times 8 + 1 \times 4)/4 = 7$, $y = (3 \times 5 + 1 \times (-3))/4 = 3$. Point **(7, 3)**.

IN WHAT RATIO DOES THE X-AXIS DIVIDE THE JOIN OF (1, −5) AND (−4, 5)?

On the x-axis $y = 0$. Let the ratio be $k : 1$: $(5k - 5)/(k + 1) = 0 \Rightarrow k = 1$. Ratio **1 : 1** (the midpoint).

SWAPPING M AND N

m goes with the **second** point (x_2, y_2) and n with the **first**. Write "A(x_1, y_1), B(x_2, y_2)" before substituting.

Quick recap

- Distance = $\sqrt{(\Delta x^2 + \Delta y^2)}$.
- Section formula (m : n): weights cross over to the far point.
- Midpoint = average of coordinates. The centroid of a triangle is the average of its three vertices.

Practice questions

1. The distance of the point $(-6, 8)$ from the origin is
   (a) 6 (b) 8 (c) 10 (d) 14
2. The midpoint of $(2, -4)$ and $(-6, 10)$ is
   (a) $(-2, 3)$ (b) $(2, -3)$ (c) $(-4, 6)$ (d) $(4, 3)$
3. The point dividing $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$ is
   (a) $(1, 3)$ (b) $(3, 1)$ (c) $(2, 3)$ (d) $(1, 4)$
4. The points $(0, 0)$, $(3, 0)$ and $(0, 4)$ form a triangle whose longest side is
   (a) 3 (b) 4 (c) 5 (d) 7
5. If $(x, 2)$ is equidistant from $(8, -2)$ and $(2, -2)$, then x equals
   (a) 3 (b) 5 (c) 4 (d) 6

Answers: 1. c 2. a 3. a 4. c 5. b